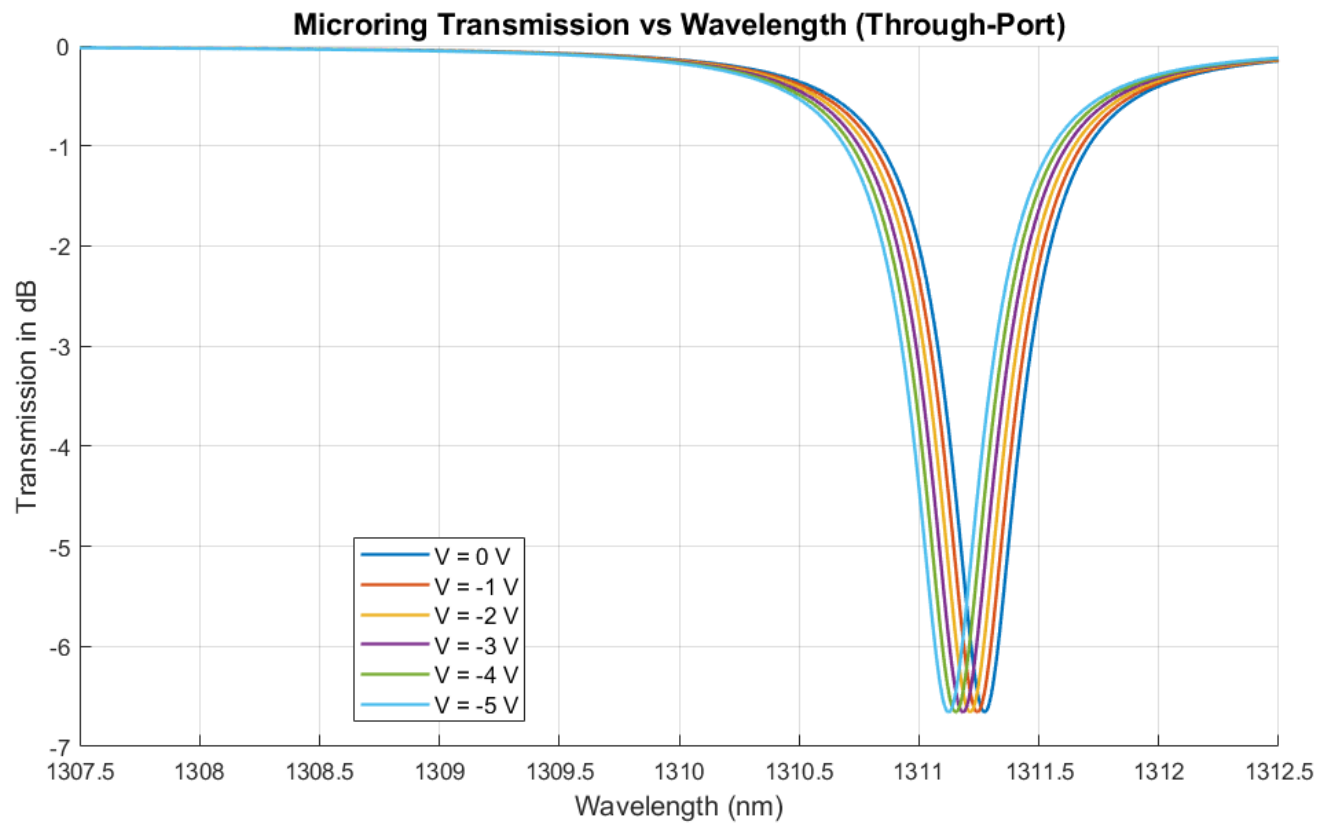




Literature review over
microring modulators.

Figure 1

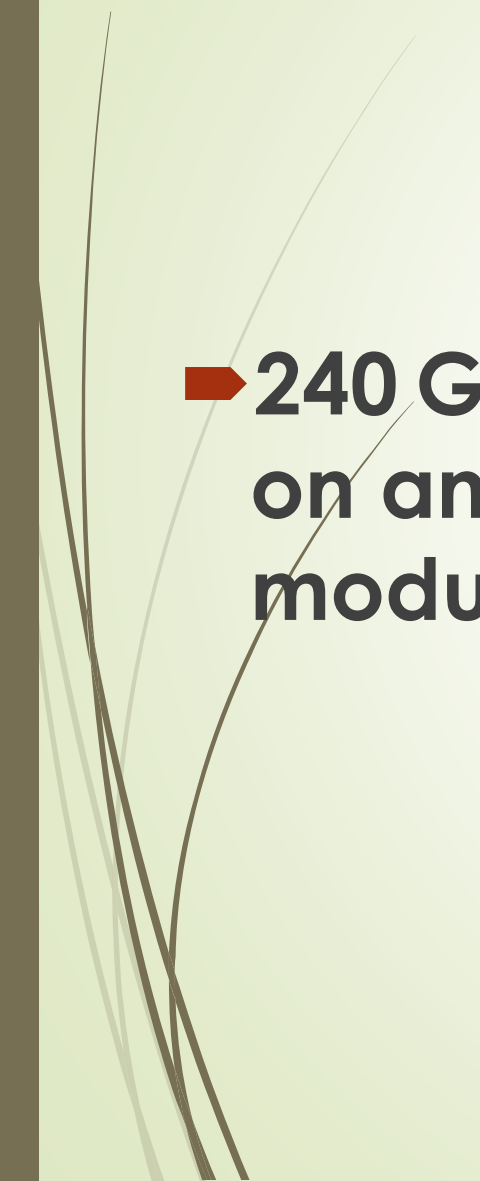
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- **240 Gb/s optical transmission based on an ultrafast silicon microring modulator**, (Zhang *et al.*, *Photonics Research*, 2022).
- **Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator Working at 100Gb/s**, (Hongjun Cai *et al.*, *IEEE PHOTONICS TECHNOLOGY LETTERS*, 2022).

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- **240 Gb/s optical transmission based on an ultrafast silicon microring modulator**

240 Gb/s optical transmission based on an ultrafast silicon microring modulator

- **Device geometry & fabrication:**

Platform: **220 nm** SOI.

Ring radius: **8 μm** (compact footprint).

Waveguide: **420 nm** width, **70 nm** slab and a Coupler gap **280 nm**.

PN junction: lateral PN with P width **230 nm**, N width **190 nm** (designed because hole concentration produces larger Δn with lower loss).

Doping: optimized to **$3 \times 10^{18} \text{ cm}^{-3}$** (P & N)

240 Gb/s optical transmission based on an ultrafast silicon microring modulator

- Device geometry & fabrication:

Measured $Q \approx 4500$

$V_{\pi} \cdot L = 0.825 \text{ V.cm}$ measured at **4.5 V** reverse bias.

optical BW $\approx 52 \text{ GHz}$ (FWHM) at resonance.

$$\frac{1}{f_{\text{mod}}^2} = \frac{1}{f_{\text{opt}}^2} + \frac{1}{f_{\text{ele}}^2}$$

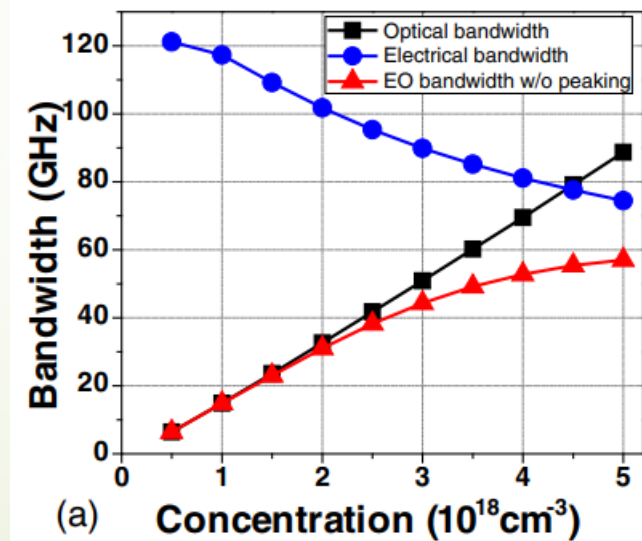
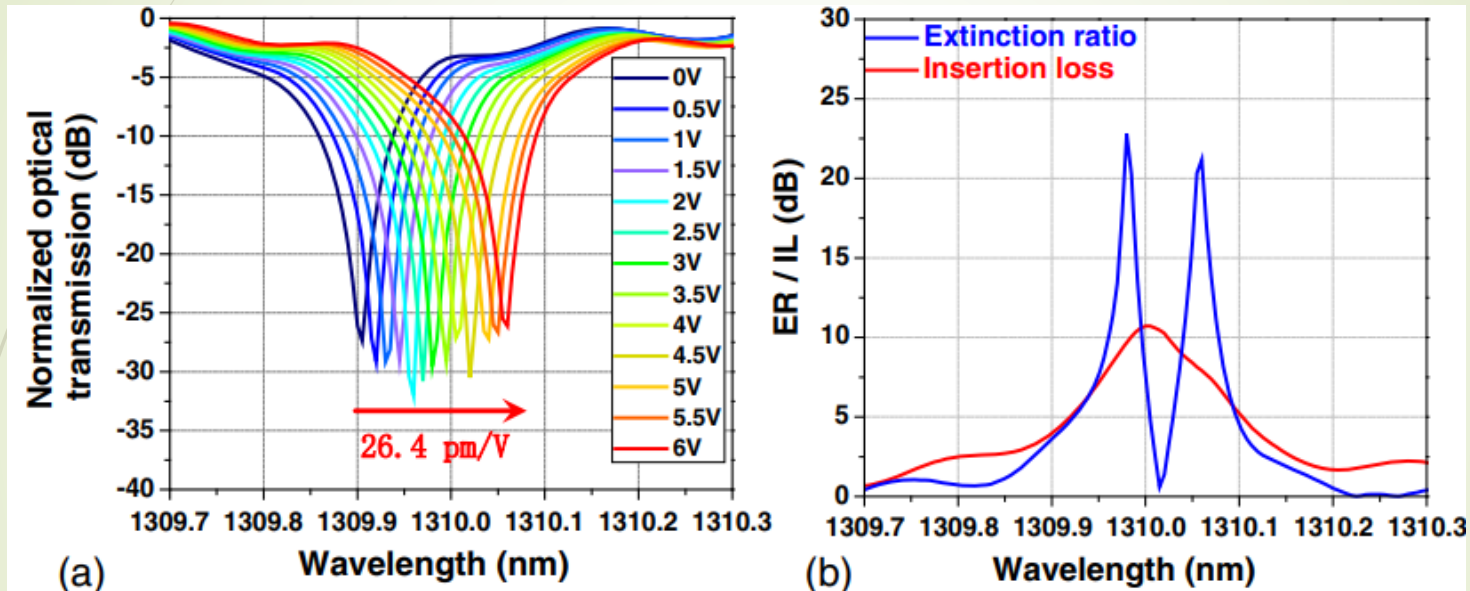


Fig.1 Simulated O bandwidth, E bandwidth, and EO bandwidth of the MRM with different doping concentrations.

240 Gb/s optical transmission based on an ultrafast silicon microring modulator



- Fig. 2(a) Normalized optical transmission by varying the reversed-bias voltages. (b) Extracted insertion loss (IL) and ER with different working wavelength with driving voltage of 3V pp.

240 Gb/s optical transmission based on an ultrafast silicon microring modulator

High-speed transmission results

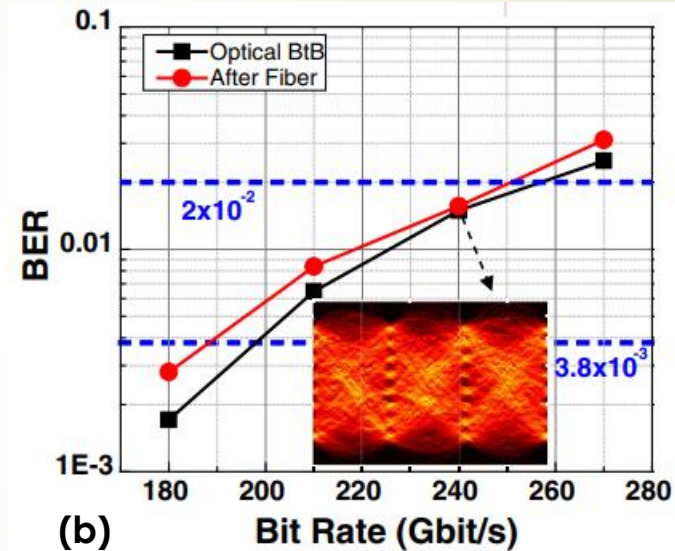
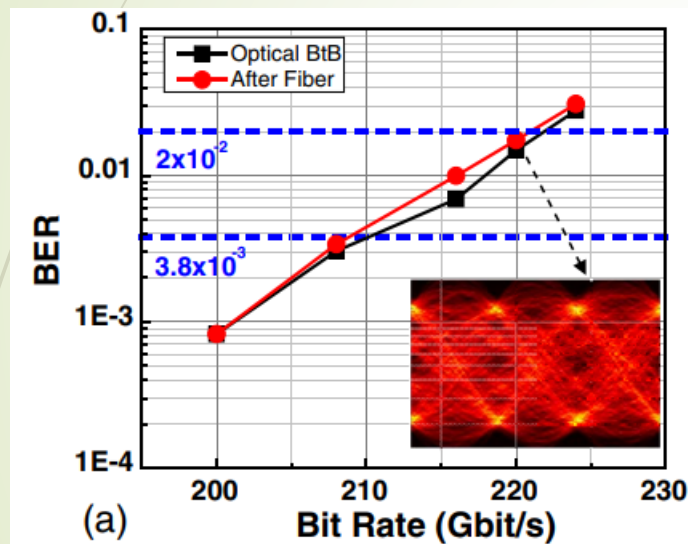


Fig. 3 Measured BER for BtB and after 2 km SSMF transmission with different data rates for (a) PAM-4 signal of the MRM and (b) PAM-8 signal of the MRM.

240 Gb/s optical transmission based on an ultrafast silicon microring modulator

120 Gb/s NRZ (BtB): BER = 5.5×10^{-4} (at 2.5 dBm received power).

220 Gb/s PAM-4 (BtB): BER = 1.5×10^{-2} ; after 2 km SSF: 1.7×10^{-2} .

240 Gb/s PAM-8 (BtB): BER = 1.4×10^{-2} ; after 2 km SSF: 1.5×10^{-2} .

240 Gb/s optical transmission based on an ultrafast silicon microring modulator

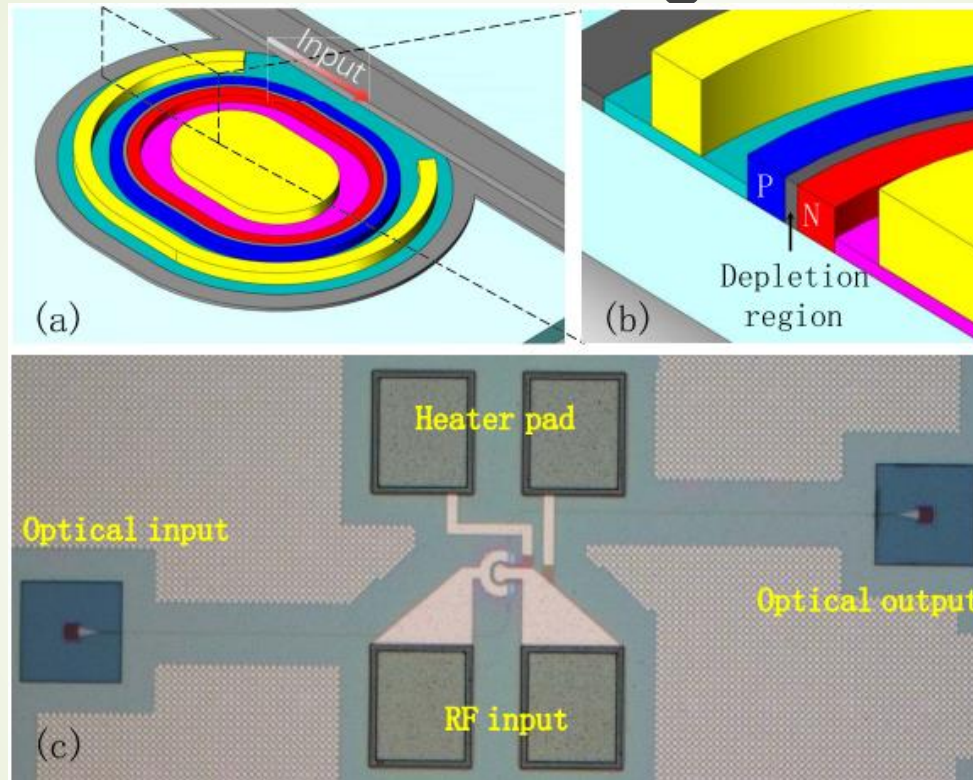
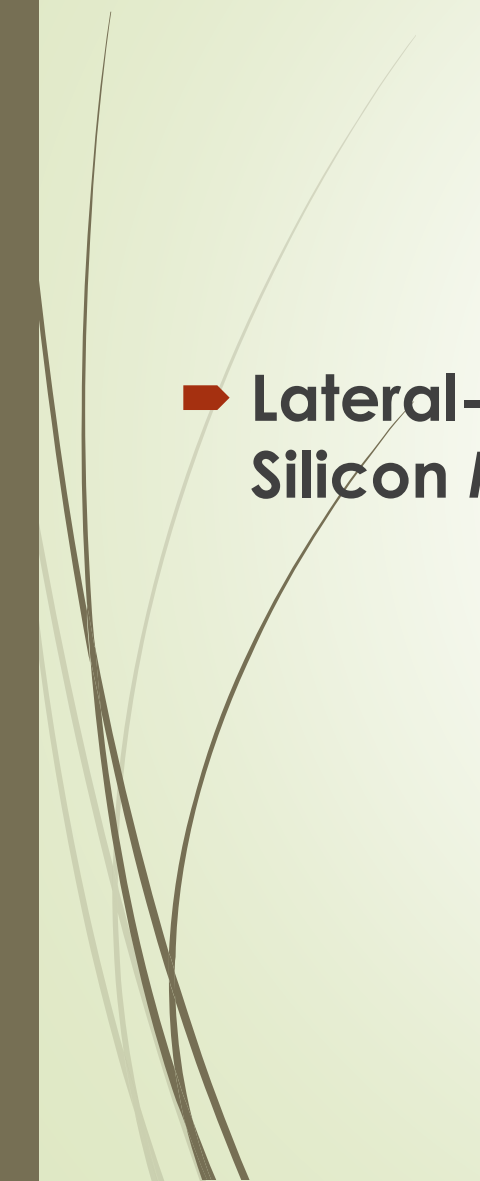


Fig. 4 (a) Schematic for the high-performance MRM where The **yellow** area is the **metal electrode** laid on top of the ring (contact metal) used for biasing and thermal tuning. (b) PN junction indicated by dashed square in (a). (c) Microscope picture of the proposed MRM.

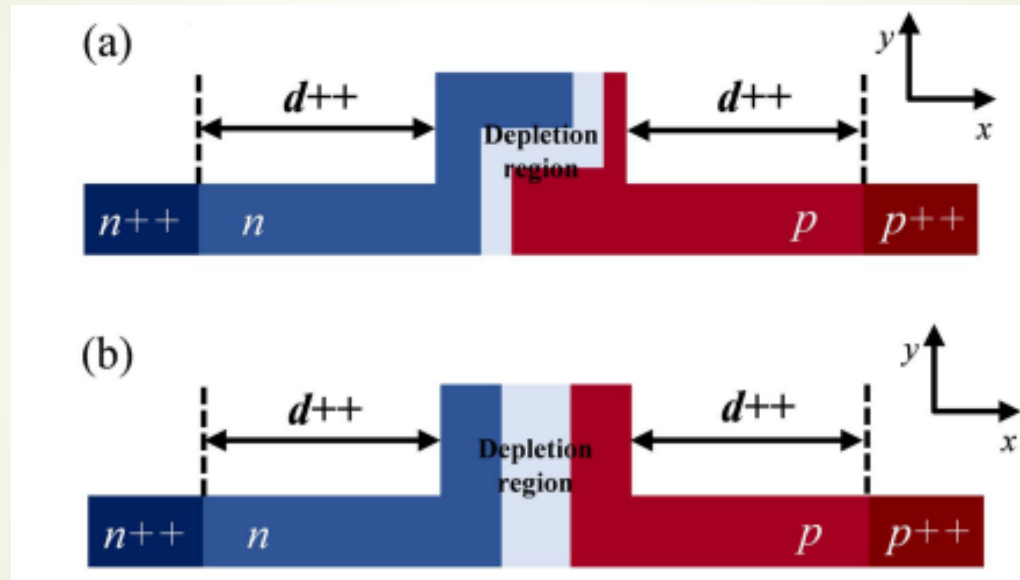
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- **Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator Working at 100Gb/s.**

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator

- The authors introduced a new lateral-zigzag PN junction designed to significantly increase modulation efficiency while keeping a high RC bandwidth and simple fabrication.
- This optimized junction is embedded into a **9- μm -**radius microring operating in both C-band and O-band, achieving up to **100 Gb/s** PAM-4.

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator

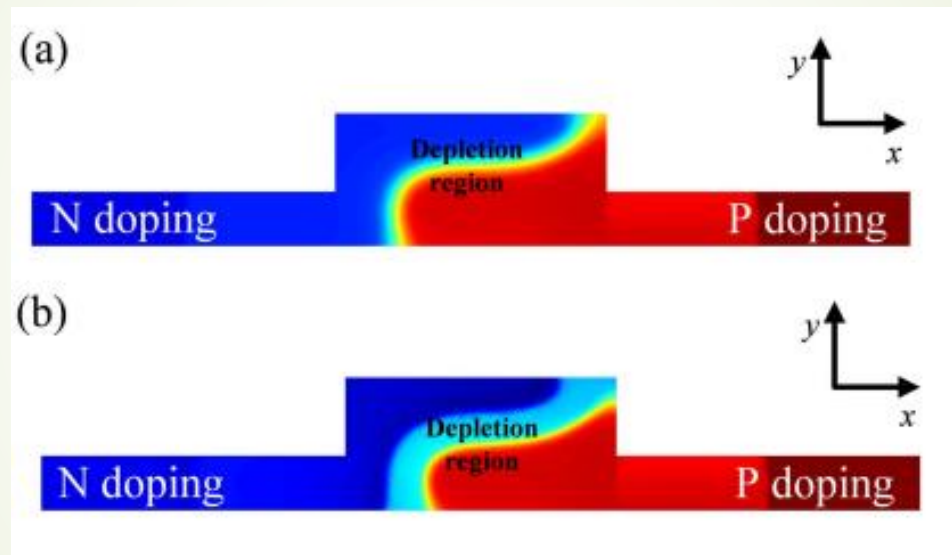
Fabrication & geometry



- Fig.5 The waveguide cross-section view of (a) the proposed lateral-zigzag PN junction and (b) the conventional lateral PN junction.

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator

Fabrication & geometry



- Fig.6 The carrier concentration profiles of the proposed lateral-zigzag PN junction at a bias voltage of (a) **0 V** and (b) **-2 V**.
- Platform: SOI, **220 nm** Si, **2 μm** BOX Waveguide: **450 nm** wide ridge, **90 nm** slab Ring, radius: **9 μm** , Coupler gap: **0.27 μm** .

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator

Fabrication & geometry

- Platform: SOI, **220 nm** Si, **2 μm** BOX Waveguide: **450 nm** wide ridge, **90 nm** slab Ring.
- radius: **9 μm** .
- Coupler gap: **0.27 μm** .
- Electro-Optic (EO) Bandwidth : **42 GHz** (-3 dB).
- $Q \approx$ **7000** .

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator

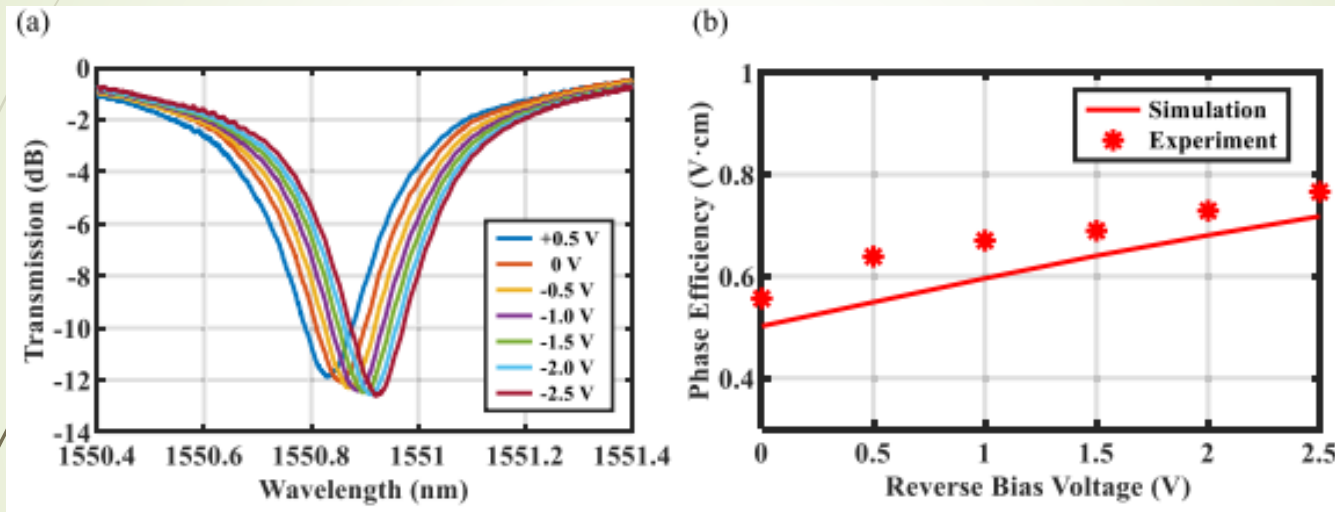


Fig.7 (a) Measured transmission spectra of the micro-ring modulator with bias voltages varying from **+0.5 V to -2.5 V**. (b) Measured (asterisks) and simulated (solid line) phase efficiency of the PN junction in the micro-ring modulator.

Lateral-Zigzag PN Junction Enabled High-Efficiency Silicon Micro-Ring Modulator

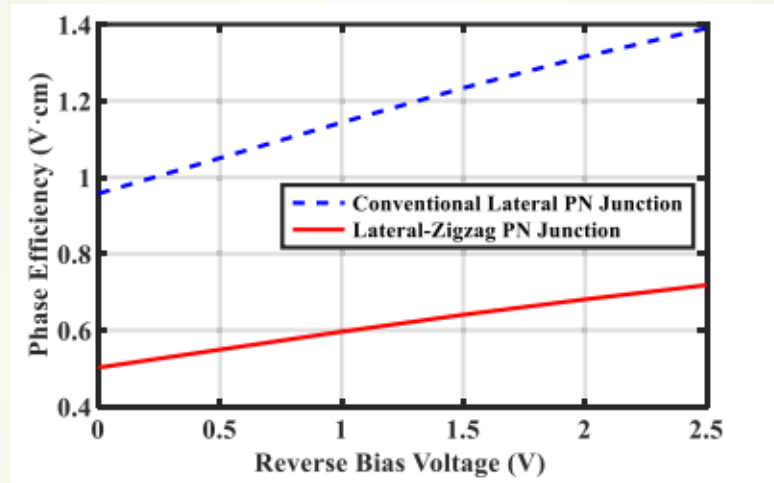


Fig.8 Simulated phase efficiency ($V_{\pi} \cdot L_{\pi}$) under different reverse bias voltage of conventional lateral PN junction and lateral-zigzag PN junction at the wavelength of **1550 nm**.



References

- Zhang, Yuguang, et al. "240 Gb/s optical transmission based on an ultrafast silicon microring modulator." *Photonics Research* 10.4 (2022): 1127-1133.
- Cai, Hongjun, et al. "Lateral-zigzag PN junction enabled high-efficiency silicon micro-ring modulator working at 100Gb/s." *IEEE Photonics Technology Letters* 34.10 (2022): 525-528.